Q.1 ..... treats the geometry of motion of bodies without taking into account their masses or the forces on them.

A. statics

B. kinematics

C. astrophysics

D. kinetics

Q.2 IF $$\vec r = 3{t^2}\hat i + 3\hat j$$ is the position of the particle at time t, then which of the following is true about the acceleration of the particle?

A. $$ a = 6t i + 3 j$$

B. $$ a = 6 i$$

C. $$ a = 6 i + 3 j$$

D. $$ a = 0$$

Q.3 If the acceleration of a particle is zero, then which of the following is true?

A. Distance traveled in each unit of time is decreasing.

B. Distance traveled in each unit of time is increasing

C. Distance traveled in each unit of time is constant

D. None of the given

Q.4 If x is the displacement of particle from a fixed point at any time t and is given by $$x = 5{t^2}$$, then which of the following will be the expression for the acceleration of the particle?

A. 10t

B. 5

C. 10

D. 5t

Q.5 If the speed "v" remains unchanged with time, then acceleration is

A. normal.

B. zero.

C. tangent.

D. constant.

Q.6 If the displacement of a particle is given by$$x = 3{t^2} + 4$$, then the acceleration of the particle at any time t will be ------.

A. 6

B. 6t

C. 0

D. 3t

Q.7 The radial component of acceleration is -----.

A. $${1}{r}{d}{{dt}}({r^2})$$

B. $$ r - r{( )^2}$$

C. none of these

D. $${1}{{{r^2}}}{d}{{dt}}({r^2})$$

Q.8 At any time t the position of a particle moving in a plane can be specified by $$(a\cos \varpi t,a\sin \varpi t)$$, where a and $$\varpi$$are constants, the normal component of velocity is ----.

A. $${{a^2}{ ^2}}$$

B. 0

C. $${a }$$

D. $$ {a }$$

Q.9 If a particle is moving with zero acceleration and it starts with velocity 10m/s, then the distance traveled in 5 seconds will be ----.

A. 15m

B. 10m

C. 60m

D. 50m

Q.10 The position of a particle moving along a path at any time t is given by $$\vec r = \sin t\cos t\hat i + {\sin ^2}t\hat j$$ and $$\theta = 7t$$. The transverse component of velocity is ----.

A. $$- 7 t$$

B. $$6 t$$

C. $$- t$$

D. $$7 t$$
